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Experiment Name: Word Frequency in Text Processing

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import gutenberg
# Download the gutenberg corpus if not already installed
nltk.download('gutenberg')
nltk.download('punkt') # Ensure the punkt tokenizer models are also
downloaded
# Load the text from the gutenberg corpus
sample = gutenberg.raw("austen-emma.txt") # Change to an existing text
from the corpus
# Tokenize the sample text
token = word_tokenize(sample)
# Create a list of the first 50 tokens
wlist = []
for i in range(50):
    wlist.append(token[i])
# Calculate the frequency of each word in the list
wordfreq = [wlist.count(w) for w in wlist]
# Print the word-frequency pairs
print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

```
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1),
('1816',
1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2),
('Emma
', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5),
('clever'
, 1), (',', 5), ('and', 3), ('rich', 1), (',', 5), ('with', 2), ('a',
1
), ('comfortable', 1), ('home', 1), ('and', 3), ('happy', 1),
('disposi
tion', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some',
1
), ('of', 2), ('the', 2), ('best', 1), ('blessings', 1), ('of', 2),
('e
xistence', 1), (';', 1), ('and', 3), ('had', 1), ('lived', 1),
('nearly
', 1), ('twenty-one', 1), ('years', 1), ('in', 1), ('the', 2),
('world'
, 1), ('with', 2)]
```